



Infection Control

Lab 9



Introduction

Infectious diseases occur as a result of invasion of micro-organisms (m.o.) into a living system or due to the actions of the products of micro-organisms or a combination of both .

Typically, the (m.o.) that may be responsible for disease in a human host are classified into four groups:

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1-Bacteria. 2-Viruses. 3-Fungi. 4-Protozoa.

The dental environment is associated with a significant risk of exposure to various m.o. , many infectious agents may be present in blood ,saliva or respiratory secretions. Infectious diseases are always serious and can be life threatening hazard to the health care workers, including dental health care workers .

Two issues need to be considered relating to this problem:

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1-Transmission of infection from **infected patients** to **dental health care workers**.

2-Transmission of infection from infected **workers** to the **general public**.

Transmission of infection in the dental operator through several means:

- 1-Direct contact** (with skin mucous membranes ,blood ,saliva or other secretions).
- 2-Indirect contact** (with contaminated instruments ,surfaces ,or operator equipment).
- 3-Airborne contaminants** (through inhalation of aerosols of oral and respiratory fluids).
- 4-Vehicle borne contaminants** (ingestion of contaminated water ,food and drugs).

Prevalent infectious conditions:

The infectious diseases which are widely prevalent and to which the dental health care workers are susceptible as follows:

- Hepatitis (B and C) .
- Herpes virus infections.
- Syphilis .
- Gonorrhea .
- Acquired Immune Deficiency Syndrome (AIDS).
- Tuberculosis

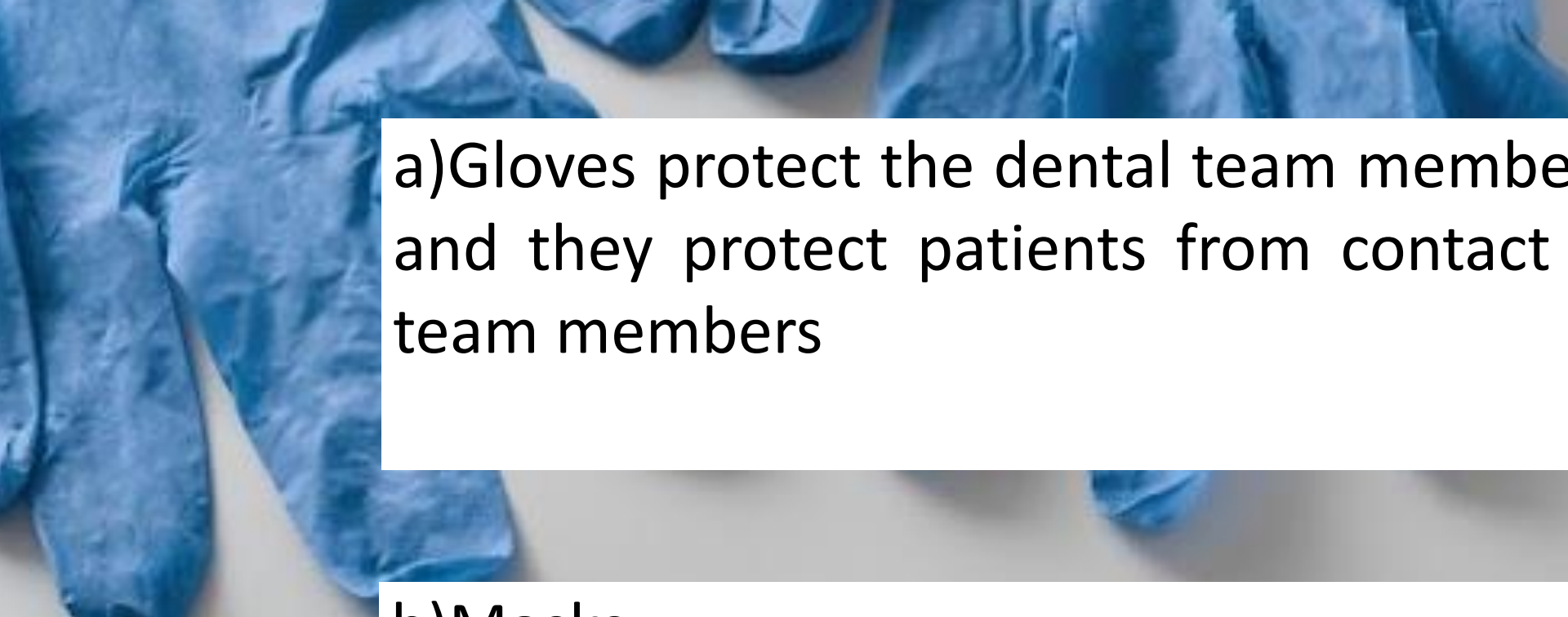
Infection control:

It is controlling the spread of disease agents by performing specific procedures. The approaches of infection control in dentistry is directed towards protection of patients and dental team from infectious diseases, the **procedure of infection control** are designed to kill or remove microbes or to protect against contamination.

Personal Protective Equipment (PPE):

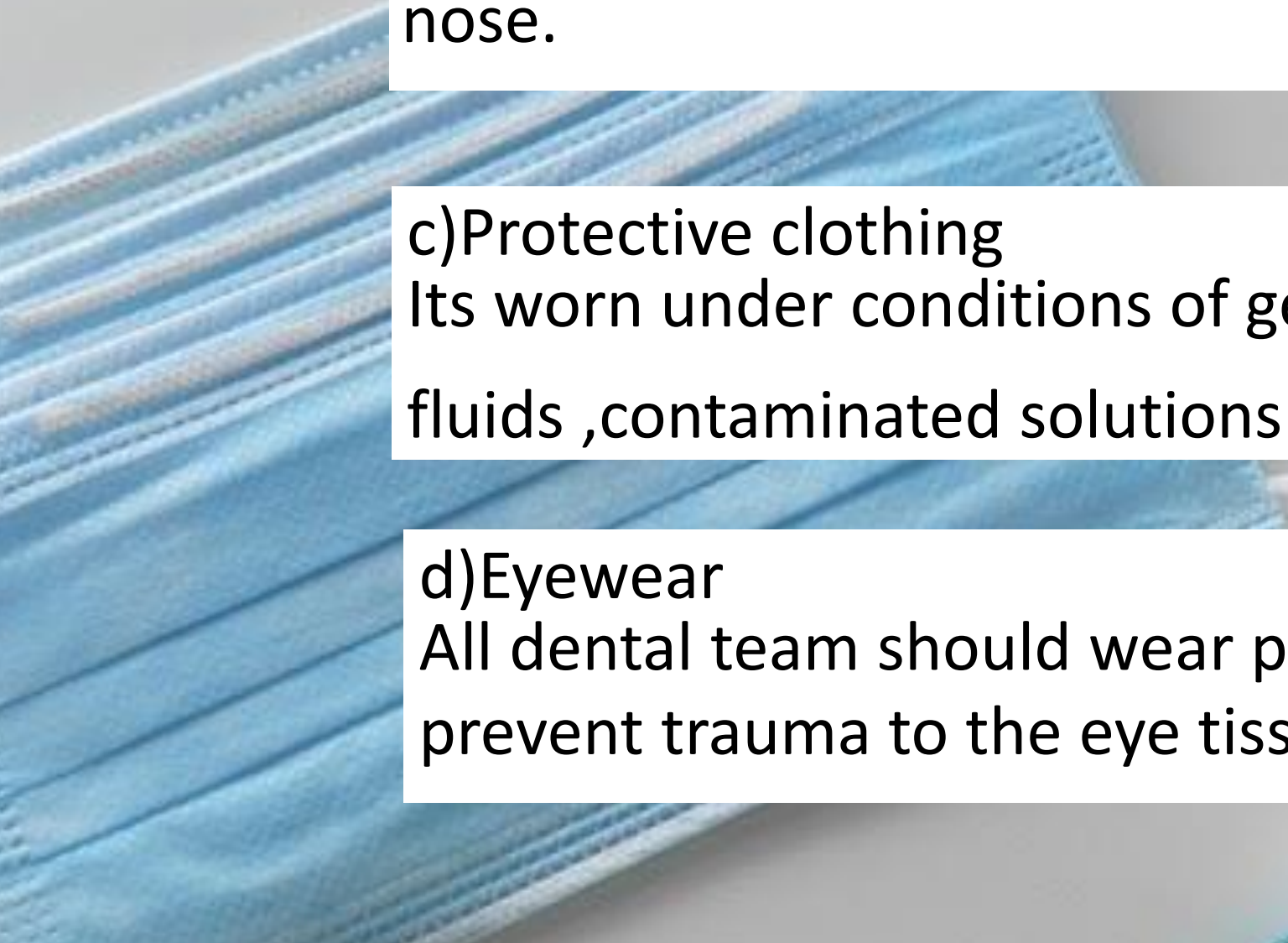
Are used in health care settings to create a barrier between health care workers and an infectious agent from the patient.

All PPE should be removed before dental health care persons (DHCP)
leave patient-care areas.



a)Gloves protect the dental team members from direct contact with patient microbes and they protect patients from contact with microbes on the hands of the dental team members

b)Masks
Face masks prevent spatter from patients mouths or splashes of contaminated solutions and chemicals from contacting the mucous membranes of the mouth and nose.



c)Protective clothing
Its worn under conditions of generating sprays, splatter, splashes or spills of body fluids ,contaminated solutions or chemicals

d)Eyewear
All dental team should wear protective eyewear in the form of glass or face shield to prevent trauma to the eye tissue from flying droplets or aerosols.

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Hand Hygiene: Proper cleaning of hands to remove dirt, organic materials, or microorganisms , Proper hand washing destroys pathogens, removes blood, debris and contaminating microorganisms and prevents accumulation of blood on damaged hand skin.

Antiseptics used for hand wash are as example: Chlorhexidine (2-4%) with 4% isopropyl alcohol, Povidone iodine (7.5 -10%).

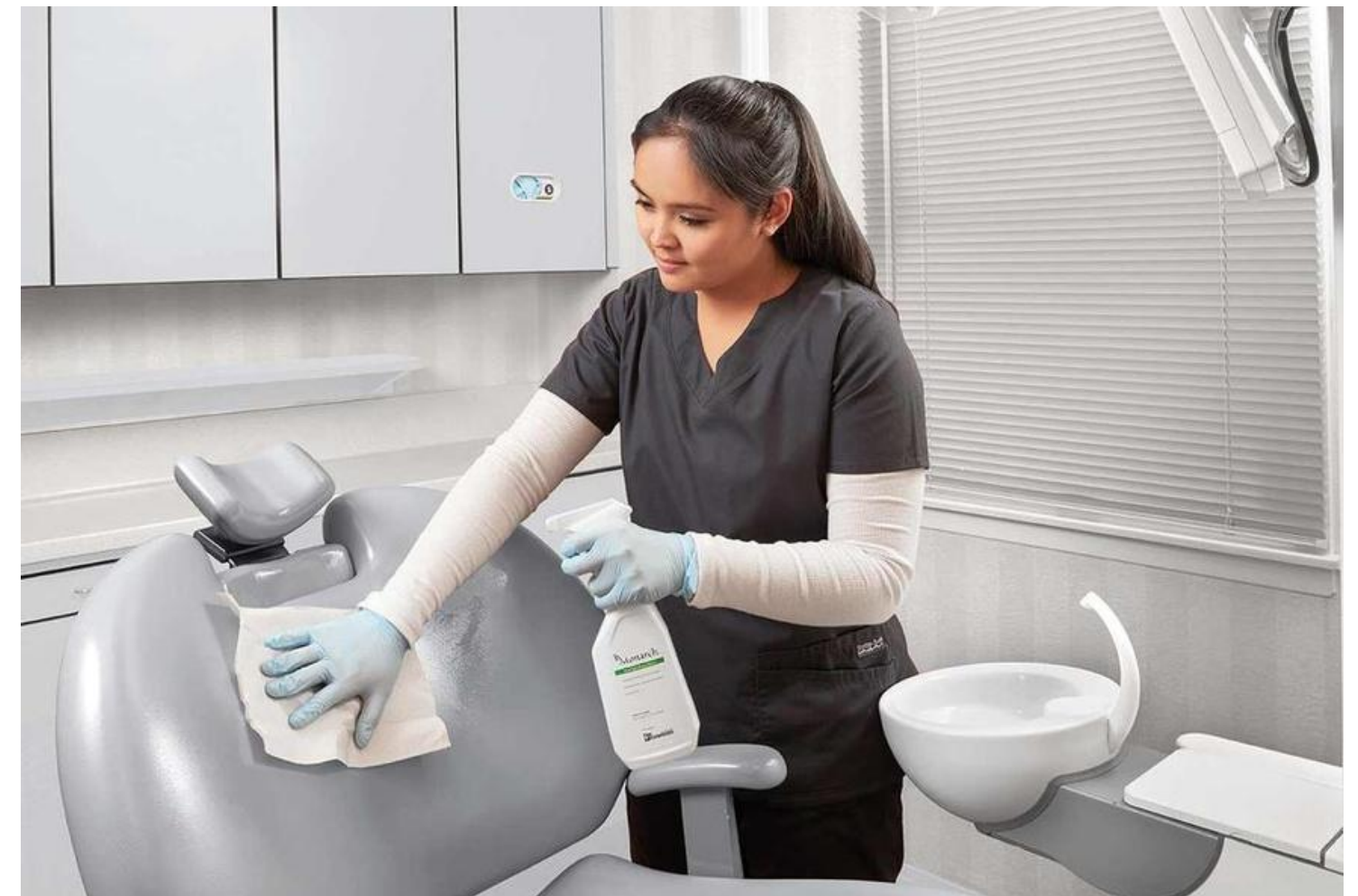
Drying is also an essential part of hand hygiene process. One should use good quality soft, disposable paper towels.

Patient Placement: Patients with known or suspected infection who pose a risk for transmission to others must be isolated. They should be placed in a single-patient room if available or with patients with the same infection. To control the source of infection when droplet or airborne transmission is suspected, it is advised to place a mask on the patient immediately.

Practice Careful Aseptic Techniques: During treatment and clean-up, blood and saliva-contaminated blood can be spread by anything that has been in patient's mouth. Practicing aseptic techniques helps in limiting the spread of blood and saliva. This is done by:

- **Careful handling of sharp instruments:** The sharp end must always point away from the hand, pass syringes with needles pointing away from anyone, pick up instruments individually.
- **Avoiding injuries:** recap dental syringes using One handed technique or Scoop technique.

All materials of dental chair should withstand repeated disinfection ,all chair functions should be controlled from foot switch to avoid contamination of hand operated switches .Common switches or latches in the chair have a number of cracks so that harbor (m.o.) , these switches should be covered by disposable covers and the underside of chair arms should be properly cleaned and disinfected.



Aerosol contamination can be minimized by:

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1. **Pretreatment mouth rinses**- 2% chlorhexidine reduces the bacterial load in dental aerosol.
2. **High volume evacuation (HVE)**- reduces contamination by more than 90%.
3. **The rubber dam**- an effective method to reduce contamination by saliva and blood.
4. **Ventilation and air filtration**- by filtering recirculated air and by introducing a substantial proportion of fresh air in each air change will substantially reduce residual contamination.



Sterilization of Instruments:

1-Pre-soaking and cleaning: Pre-soaking of contaminated instruments keeps them wet until a thorough cleaning can occur. This procedure prevents blood and saliva from drying on the instruments and facilitates cleaning of instrument which is achieved by

(a) **Hand scrubbing of contaminated instruments.**

(b) **Ultrasonic cleaning is a mechanical cleaning system** that reduces handling of contaminated instruments and has been shown to be effective in removing dried blood and saliva.

2) Packaging

After cleaned instruments have been rinsed and dried , they are to be packaged before sterilization. This packaging protects the inst. from becoming contaminated after sterilization and before use at chairside .



3) Sterilization

Heat sterilization

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The common heat sterilizing methods available for dental office are:

- **Steam** 20-30 min .at 121 °C or 2-10 min. at 134 °C .
- **Moist heat (autoclave)** 15-30 min .at 121 -132 °C .
- **Chemical vapor** (Chemiclave) 20-40 min. at 134 °C.
- **Dry heat** (oven type) 2 hrs. at 160°C . or 170 for 1 hour.



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- **Sterilization**: It is a process of removing or killing all viable microorganism including large number of resistant bacterial spores using physical and chemical procedure.
- **Disinfection**: It is a process of removing or killing most, but not all, viable organism (e.g bacterial spores) using physical & chemical procedure.

Some Infectious Diseases of Concern to Dental Practitioners:

Hepatitis B (HBV)

- It is critical that healthcare workers who are at high risk of percutaneous exposure to HBV **get vaccinated**. This includes those in the dental practice. Healthcare workers should also receive **post vaccination testing** to ensure the presence of anti-HBs, therefore confirming their immunity.

Hepatitis C Virus

- All health care persons should adhere to the appropriate standard precautions by using personal protective equipment, to avoid both exposure to the patient and to the healthcare practitioner. **Transmission via blood splatter has been reported.**

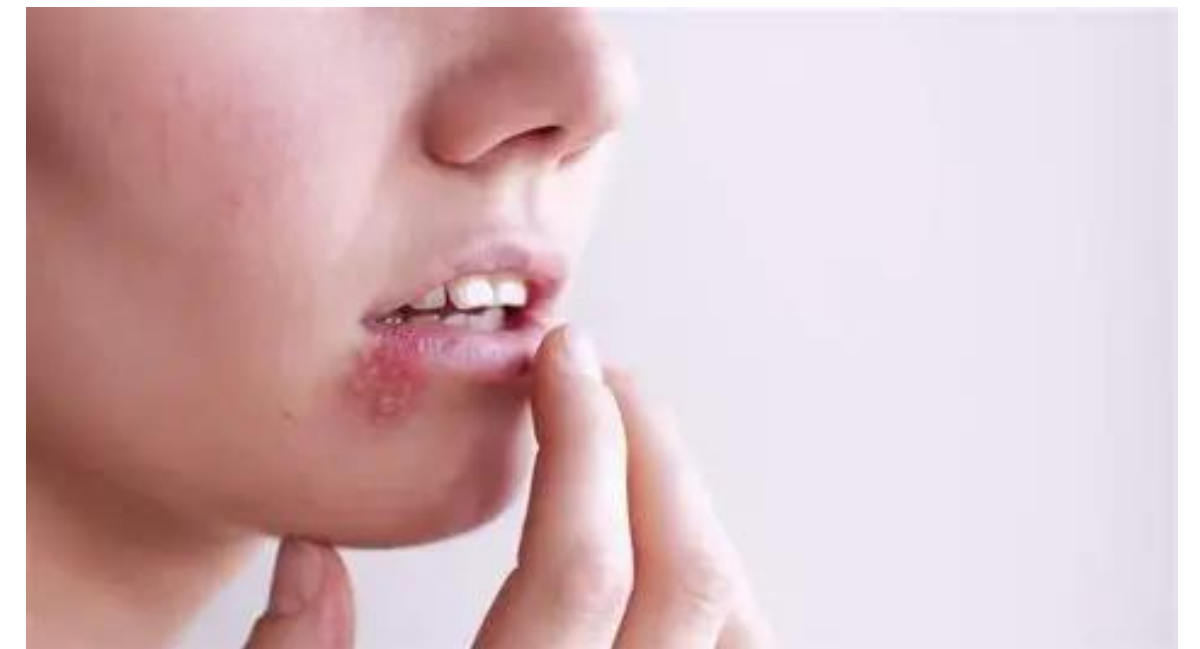
Herpes Simplex Virus (HSV)

- The Herpes Simplex Virus (HSV) is often categorized into two distinct types: HSV-1 and HSV-2.

Transmission occurs through **direct contact with a lesion, mucosal surface, or the secretions (oral or genital) of an infected host.**

Recurrent HSV-1 infection

Infected individuals with active symptoms of oral herpes should avoid oral contact with others and avoid sharing objects that have possibly been in contact with **saliva.**





**Thanks
For listening**

